

1 Potential Flow Equation in Two-Dimensions

In this homework you will use finite element methods to examine the linear potential flow around NACA 4-digit airfoils shape. While the potential flow approximation greatly simplifies fluid flow behavior (which is described more fully by the Navier-Stokes equations), this potential flow model has been instrumental in the development and design of aircraft as well as other high performance vehicles.

We provide several files for you to use. One of which is a very simple mesh generator to create NACA 4-series half domain airfoil meshes. The domain in this case is the region in which the potential flow will exist above an airfoil. An example two-dimensional geometry for a NACA 0045 is shown below:

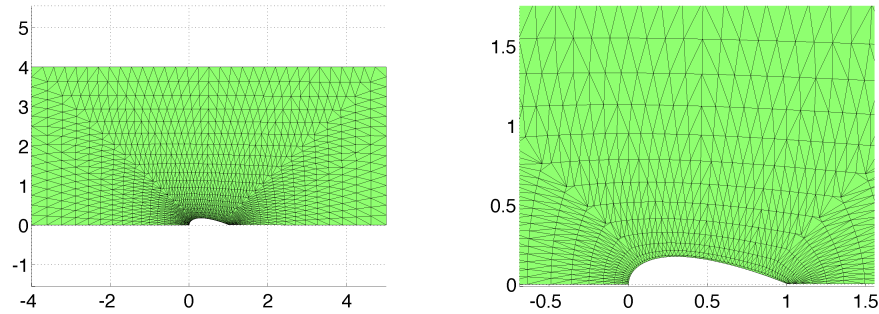


Figure 1: (left) The complete discrete geometry (right) zoom of the geometry in question)

2 Theory

You will use potential flow theory to explore the flow around the various objects in question. Consider the PDE that describes the conservation of mass for an incompressible fluid:

$$\nabla \cdot \vec{u} = 0. \tag{1}$$

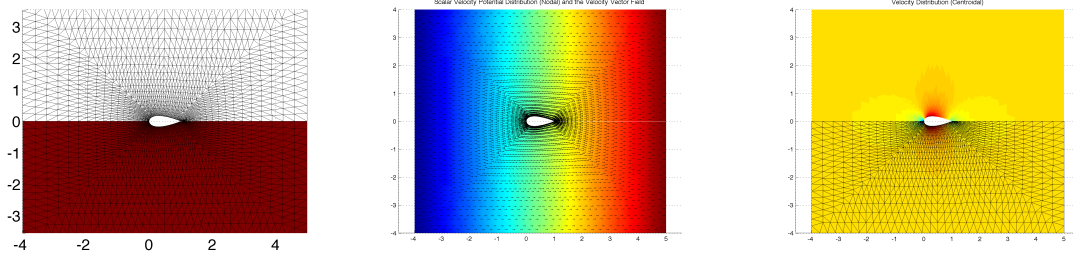


Figure 2: (left) The mirrored domain (middle) the scalar potential (right) the flow velocity)

We can introduce a scalar potential function, ϕ , such that:

$$\nabla \phi = \vec{u} \quad (2)$$

the flow described using this equations is irrotational, meaning the potential flow will have the property of irrotationality (a way to think of this is that locally the flow does not “spin”):

$$\nabla \times \vec{u} = 0 \quad (3)$$

The equation for the scalar potential (equation 2) can be substituted in as the velocity in the conservation of mass equation (equation 1) to yield:

$$\begin{aligned} \nabla \cdot \vec{u}(x, y) &= 0 \\ \nabla \cdot \underbrace{(\nabla \phi(x, y))}_{\text{Velocity from potential flow}} &= 0 \\ \nabla^2 \phi(x, y) &= 0 \\ \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} &= 0 \end{aligned}$$

The potential flow equation:

$$\boxed{\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0} \quad (4)$$

will be solved using FEM in this homework project. The domain will be the half-domain, representing the flow above the airfoil.

2.1 Boundary Conditions

The upper and lower boundaries (top and bottom boundaries) have a prescribed no-normal flow boundary condition. This includes the airfoil (or any other aerodynamic) shape being

examined. This corresponds to the following boundary condition expression:

$$\boxed{\vec{u} \cdot \hat{n} = \frac{\partial \phi}{\partial n} = 0 \quad \text{on upper and lower boundaries.}} \quad (5)$$

At the left and right hand sides, we will apply a prescribed potential value. This will setup the necessary gradient in the potential to mimic a freestream flow around the airfoil.

Based on a unit freestream velocity and the relevant potential flow theory, we will set the value at these boundaries equal to the value of the **x-coordinate** at those boundary node locations.

This will setup a uniform freestream flow entering and exiting the domain:

$$\boxed{\phi(x = X_{LHS}, y) = X_{LHS} \quad \text{on left boundary}} \quad (6)$$

$$\boxed{\phi(x = X_{RHS}, y) = X_{RHS} \quad \text{on right boundary}} \quad (7)$$

These boundary conditions should be relatively easy to implement in the code because each node's (x, y) coordinates are easily determined based on the node number and the matrix of nodal coordinates. To make this easier, the provided geometry routine hands back the locations where the Dirichlet condition will be applied (**BIG hint:** it only hands back these locations – so you need only find and assign the x-coordinate value as a boundary condition for each).

3 Skeleton Code: MATLAB files provided for this homework

In order to support your completion of this homework, a series of (semi-complete) skeleton codes has been provided:

```
FEMAirfoilSolver_Skeleton.m\\
getDiscreteGeometry.m\\
calculateArea.m\\
getNACA.m
```

You will need to modify the following code to get this the solver to a working state:

```
FEMAirfoilSolver_Skeleton.m
```

GOOD LUCK!!!

4 Tasks and Deliverables

1. **(2.5 points)** Starting from the strong form of the partial differential equation:

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0, \quad (8)$$

derive the weak form of the equation that will be used as the basis of the finite element method. Show and describe all steps.

2. **(2.5 points)** Describe the difference/benefits/drawbacks of the weak vs. strong form of the equations? Suggest using a table for this question.
3. **(2.5 points)** In your own words, describe how the FEM (and FVM) is different than the finite difference methods we examined in the first 1/3 of the course. You will want to focus on how these methods represent the solution.
4. **(2.5 points)** You will use linear-hat nodal basis functions to represent the solution and the test functions in your solver. Describe/draw how these linear hat basis functions look in **two** dimensions for:
 - (a) For an individual node of the mesh i in the triangulation, show the "tent/hat" basis function (nodal hat basis function)
 - (b) For an individual triangular element k , show the three linear basis functions on a triangular element.
5. **(10 points)** For a 2D problem, write the discrete form of the weak form of the governing equation for a weighting function i for **a node in the domain**. Explain how this equation represents one of the many equations that will form the linear system of equations for this problem.
6. **(10 points)** For a 2D problem, write the discrete form of the equation for a node i on the **boundary of the domain**. Show/explain how you will handle Dirichlet vs. Neumann boundary conditions.
7. **(20 points)** For this 2D problem, derive and how the elemental matrices (so that each element or triangle has its own matrix) that will be used to build the overall A -matrix (stiffness matrix). Also show/derive the elemental load vectors that will be used to construct the RHS. Be sure to illustrate how you will calculate the gradient of the basis function values using the equation for a plane surface.
8. **(10 points)** Describe the Stamping procedure that will be used to construct the global matrix using the elemental matrices. You may use both words and example matrices. For each boundary condition, describe the implementation in your FEM code.

9. **(30 points)** Add the elemental matrices and boundary conditions to the provided pseudo code and complete the FEM solver so that it works (currently the solver has some incorrect numbers in it to allow it to "work" when you press run). You may optionally develop your own two-dimensional FEM solver. If you choose to use the MATLAB version provided, please read through and understand the comments and structure. If you choose to develop your own solver, please use the geometry definition tool provided to generate base grids. Solve the potential flow problem for a cylinder (defined in the geometry definition by setting the shape as follows:

`Shape = 1;`

and plot the scalar potential using a minimum grid refinement value of 3.

10. **(5 points)** Show how the value of your scalar potential solution at the front of the object (location 0,0) converges with grid refinement.
11. **(5 points)** Application problem. Assume that you need to design a fairing for a sensor and the associated wiring that will be installed perpendicular to the flow on a car's roof (because the fairing needs to contain the sensor and wires, it will need a consistent height).

Examine several different half-airfoils to determine how the velocity distribution near the sensor differs for each. An airfoil can be defined by changing the "shape" defined in the code; for a **NACA 0025** airfoil, the shape is defined as:

`Shape = [1 0025]`

Based on your exploration, make a recommendation for the design of such a fairing. Note: if you feel adventurous, you could consider exploring non-airfoil shapes.